

Assignment-2 Conditional

Q1 What is a conditional statement in programming?

Ans. A conditional statement is a programming construct that executes specific blocks of code only if a defined boolean condition evaluates to true. It acts as a decision-making control flow, allowing programs to react dynamically to different inputs or states.

Q2 What does an if statement do?

Ans An if statement evaluates a condition and executes a specific block of code only if that condition is true.

Q3 What is the purpose of an else block?

Ans An else block provides a fallback mechanism that executes code only when the condition in the preceding if statement evaluates to false. It ensures that your program has a defined action for every scenario, preventing the logic from simply doing nothing when a specific condition isn't met.

Q4 When is an else block executed?

Ans An else block is executed exclusively when the condition specified in its corresponding if statement (and any preceding else if statements) evaluates to false. It acts as a final catch-all container to handle any scenarios that were not covered by the initial conditions

Q5. What is a Boolean expression?

Ans A Boolean expression is a logical statement that evaluates exclusively to either true or false. These expressions are the foundation of decision-making in programming, often using comparison operators (like <, >, ==) or logical operators (like &&, ||, !) to determine the state of data.

Q6 Write the syntax of a simple if statement.

Ans if (condition) {

 // Code to execute if the condition is true

}

Q7 What operator is commonly used to test equality?

Ans The **equality operator** (==) is the standard tool used to test whether two values are equal. It compares the operands on both sides and returns true if they are identical and false if they are not

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int a = 5;
```

```
    int b = 5;
```

```

if (a == b) {
    cout << "Values are equal.";
}

return 0;
}

```

Q8. What does the modulus (%) operator return?

Ans . The modulus operator returns the remainder of a division operation between two numbers. For example, 10 % 3 returns 1 because 3 goes into 10 three times with a remainder of 1.

Q9. How can you check if a number is even?

Ans . You can check if a number is even by using the modulus operator to see if the remainder is 0 when divided by 2 (e.g., if (number % 2 == 0)). If this evaluates to true, the number is even.

```

#include <iostream>
using namespace std;

int main() {
    int num = 10;
    if (num % 2 == 0) {
        cout << num << " is even.";
    }
    return 0;
}

```

Q10. How can you check if a number is odd?

Ans . You can check if a number is odd by using the modulus operator to see if the remainder is not 0 when divided by 2 (e.g., if (number % 2 != 0)). Alternatively, checking if number % 2 == 1 also confirms it is odd.

Program to chake number is odd or even:

```

#include <iostream>
using namespace std;

int main() {
    int num = 7;
    if (num % 2 != 0) {
        cout << num << " is odd.";
    }
    return 0;
}

```

}

Q11. What is the purpose of else-if?

Ans . The else-if statement allows you to check multiple sequential conditions if the initial if condition is false. It provides a way to handle several specific scenarios in a chain rather than just one alternative (e.g., grading scales: if score is 90, else if score is 80, etc.).

Q12. What is a nested if statement?

Ans . A nested if statement is an if block placed inside another if block. It is used when you need to perform a secondary check only after the primary condition has already been met.

Q13. Can an if statement exist without an else?

Ans 13. Yes, an if statement can exist entirely on its own. In this case, the program simply executes the code inside the if block if the condition is true; if the condition is false, it skips the block and continues with the rest of the program without taking any action.

Q14. What is the output when a false condition is tested in an if statement without else?

Ans . There is no output from the if block because it is skipped entirely. The program flow simply proceeds to the very next line of code following the closing brace of the if statement.

Q15. What is the role of braces {} in C/C++ conditionals?

Ans . Braces define a "block" of code, grouping multiple statements together so that they are treated as a single unit by the if or else condition. Without braces, only the very next single statement after the condition is considered part of that conditional logic.

Q16. What does > mean?

Ans . The > is the "greater than" comparison operator. It returns true if the value on the left is strictly larger than the value on the right, and false otherwise (e.g., 10 > 5 is true).

Q17. What does < mean?

Ans . The < is the "less than" comparison operator. It returns true if the value on the left is strictly smaller than the value on the right, and false otherwise (e.g., 3 < 8 is true).

Q18. What does >= mean?

Ans . The >= is the "greater than or equal to" comparison operator. It returns true if the value on the left is either larger than or exactly equal to the value on the right.

Q19. What does <= mean?

Ans . The <= is the "less than or equal to" comparison operator. It returns true if the value on the left is either smaller than or exactly equal to the value on the right.

Q20. What does != mean?

Ans . The != is the "not equal to" comparison operator. It returns true if the two values being compared are different from each other.

Q21. What is a switch statement?

Ans . A switch statement is a control flow structure that allows a program to test a variable against a list of constant values. It is often a cleaner, more readable alternative to using a long chain of if-else if statements when checking a single variable against many possible discrete outcomes.

Q22. What is a case label in a switch statement?

Ans . A case label represents a specific value that the switch expression is compared against. If the expression matches the value defined in a case label, the program executes the code block associated with that label.

Q23. Why is break used in switch?

Ans . The break statement is used to exit the switch block immediately after a matching case is executed. Without a break, the program would continue to execute the code in all subsequent case blocks (a behavior known as "fall-through"), which is usually unintended.

Q24. What happens if break is omitted?

Ans . If break is omitted, "fall-through" occurs, meaning the program continues executing the code in the subsequent case blocks regardless of whether those cases match the expression. This continues until a break is encountered or the end of the switch block is reached.

Q25. What is the default case in switch?

Ans The default case is an optional block that executes when the switch expression does not match any of the provided case labels. It serves as a catch-all safety net, similar to the final else block in an if-else chain.

Q26. Can switch compare strings in C++?

Ans . No, standard C++ switch statements can only evaluate integral types (such as int, char, or enum). You cannot directly use strings in a switch case, as strings are not considered integral types in the C++ language.

Q27. Which statement is better for many constant choices: if-else or switch?

Ans . The switch statement is generally better for many constant choices because it is often more readable and can be optimized by the compiler into a jump table. This makes it more efficient than an if-else ladder, which must evaluate every condition sequentially.

Q28. How do you test whether a value is positive?

Ans You test if a value is positive by comparing it to zero using the "greater than" operator (>). For example, if (number > 0) returns true if the value is strictly greater than zero.

Program :

```
#include <iostream>

using namespace std;

int main() {

    int num = 15;
```

```
if (num > 0) {  
    cout << num << " is a positive number."  
}  
  
return 0;  
}
```

Q29. How do you test whether a value is negative?

Ans . You test if a value is negative by using the "less than" operator (<) to check if the number is strictly less than zero.

Example:

```
if (num < 0) {  
    cout << num << "it is a negative number."  
}
```

Q30. How do you test whether a value is zero?

Ans . You test if a value is exactly zero by using the equality operator (==) to check if the number equals zero.

Example:

```
if (num == 0) {  
    cout << "The value is zero."  
}
```

Q31. Write a condition to check if age is at least 18.

Ans . To check if an age is "at least 18," you use the "greater than or equal to" operator (>=) to verify that the value is 18 or higher.

Example:

```
int age = 20;  
  
if (age >= 18) {  
    cout << "You are an adult."  
}
```

Q32. Write a condition to check if marks are above 50.

Ans . To check if marks are strictly above 50, use the greater than operator (>).

```
if (marks > 50) {  
    cout << "Passed";  
}
```

Q33. What is decision making in programming?

Ans . Decision making is the ability of a program to execute different segments of code based on specific conditions. It allows software to behave dynamically by evaluating data and choosing which path of logic to follow.

Q34. Can nested if statements have multiple levels?

Ans . Yes, nested if statements can have multiple levels, allowing for complex decision trees where secondary or tertiary conditions are evaluated only if all parent conditions are met.

C++ example :

```
if (itemInStock) { // Level 1
    if (userBalance >= price) { // Level 2
        if (hasValidCoupon) { // Level 3
            cout << "Purchase successful with discount!";
        } else {
            cout << "Purchase successful at full price.";
        }
    } else {
        cout << "Insufficient balance.";
    }
} else {
    cout << "Item is currently out of stock.";
}
```

Q35. What is logical AND (&&)?

Ans . The logical AND (&&) operator evaluates to true only if **both** connected conditions are true. If either side is false, the entire expression results in false.

// Example: Both conditions must be met

```
if (age >= 18 && hasTicket == true) {
    cout << "Entry allowed.";
}
```

Q36. What is logical OR (||)?

Ans . The logical OR (||) operator evaluates to true if **at least one** of the connected conditions is true. The expression only results in false if both sides are false.

```
if (isWeekend || isHoliday) {  
    cout << "You can sleep in!";  
}
```

Q37. What is logical NOT (!)?

Ans . The logical NOT (!) is a unary operator that inverts the boolean value of an expression. It turns true into false and false into true.

Example:

```
if (!isLoggedIn) {  
    cout << "User is active.";  
}
```

Q38. How do you check if a year is divisible by 4?

Ans You check this using the modulus operator (%) by checking if the remainder when dividing by 4 is zero

```
if (year % 4 == 0) {  
    cout << "Year is divisible by 4.";  
}
```

Q39. What is a leap year?

Ans . A leap year is a calendar year that contains an additional day (February 29th) to keep the calendar year synchronized with the astronomical year. It occurs every four years, with specific exceptions for century years.

Q40. What is the first condition for a leap year?

Ans . The first condition is that the year must be evenly divisible by 4. If a year is not divisible by 4, it is immediately confirmed not to be a leap year.

Q41. How do you find the largest of two numbers?

Ans 1. Problem Statement

Write a program to find the largest of two numbers using an if-else statement.

2. Algorithm / Steps

Start.

Declare and initialize two numbers.

Compare the two numbers.

If the first number is greater, print it.

Otherwise, print the second number.

Stop.

3. Dry Run with Sample Input

Assume:

num1 = 25

num2 = 18

step	condition	result
1	25>18	true
2	print 25	Largest number=25

Program :

```
#include<iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int num1 = 25, num2 = 18;
```

```
    if(num1 > num2)
```

```
        cout << "Largest Number = " << num1;
```

```
    else
```

```
        cout << "Largest Number = " << num2;
```

```
    return 0;
```

```
}
```

Output -

```
Largest Number = 25
```

6. Explanation of the Logic Used

The program compares the two numbers using the > operator. If the first number is greater, it is displayed; otherwise, the second number is displayed.

Q42. How do you find the largest of three numbers?

Ans 1. Problem Statement

Write a program to find the largest among three numbers using if-else statements.

2. Algorithm / Steps

Start.

Declare and initialize three numbers.

Compare the first number with the other two.

If it is largest, print it.

Otherwise, compare the second and third numbers.

Print the largest number.

Stop.

3. Dry Run with Sample Input

Assume:

a = 15

b = 42

c = 30

step	condition	result
1	15>42&&15>30	false
2	42>30	true
3	print 42	Largest 42

Program :

```
#include<iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int a = 15, b = 42, c = 30;
```

```
    if(a > b && a > c)
```

```
        cout << "Largest Number = " << a;
```

```

else if(b > c)

    cout << "Largest Number = " << b;

else

    cout << "Largest Number = " << c;


return 0;

}

```

Output :

Largest Number = 42

=== Code Execution Successful ===

6. Explanation of the Logic Used

The program first checks whether the first number is greater than both remaining numbers. If not, it compares the second and third numbers to determine the largest value.

Q43. What is an if-else-if ladder?

Ans An if-else-if ladder is a sequence of conditions checked one after another. The first true condition executes its block, and the remaining conditions are skipped.

Q44. What is the difference between if and switch?

Ans The if statement is used when conditions involve relational operators (>, <, >=, etc.) or logical operators (&&, |). It is more flexible and can handle complex expressions.

The switch statement is used when there are multiple constant choices for a single expression.

It checks only for equality and makes the code simpler and easier to read when dealing with many fixed options.

Q45. Can switch use ranges directly?

Ans No. A switch statement cannot use ranges directly in C/C++. It only works with constant case values.

Q46. What is fall-through in switch?

Ans Fall-through occurs when the break statement is omitted, causing execution to continue into the next case.

Q47. What data type is usually used for conditions?

Ans The Boolean (bool) data type is usually used for conditions because it stores either true or false.

Q48. Why are conditional statements important?

Ans Conditional statements are important because they allow programs to make decisions and perform different actions based on different situations.

Q49. Give one real-life example of a conditional statement.**Ans 1. Problem Statement**

Write a program representing a real-life example of a conditional statement by checking voting eligibility.

2. Algorithm / Steps

Start.

Declare and initialize age.

Check whether age is 18 or above.

If yes, display eligible for voting.

Otherwise, display not eligible.

Stop.

3. Dry Run with Sample Input

Assume:

age = 20

step	condition	result
1	20 >= 18	true
2	print eligible	done

Program :

```
#include<iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int age = 20;
```

```
    if(age >= 18)
```

```
        cout << "Eligible for Voting";
```

```
    else
```

```
cout << "Not Eligible for Voting";  
  
return 0;  
}
```

Output :

Eligible for Voting

=== Code Execution Successful ===

6. Explanation of the Logic Used

This program demonstrates a real-life conditional statement. A person's age is checked to determine whether they satisfy the condition required for voting eligibility. If the age is 18 or more, the person can vote; otherwise, they cannot.

Q50. Name four conditional structures covered.

Ans The four conditional structures covered are:

if

if-else

if-else-if ladder

switch